GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Data Sheets for Line Pipes**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-001
Document Revision	R01
Document Status	Issued For Review
Originator	S. Ikechukwu
Security Classification	Restricted
Issue Date	**22 Jul 2022**

Data Sheets for Line Pipes

R01	22.07.22	Issued for Review	S.Ikechukwu	E.Nwachukwu	A. Duru	
R00	22.07.22	Issued for Interdisciplinary Check	S.Ikechukwu	E.Nwachukwu	A. Duru	
Rev #	**Issue Date**	**Status Description**	**Originator**	**Reviewer**	**Approver (Epic)**	**Approver (BFPCL)**

Revision History

Rev No.	Date	Description of Revision
R00	22/07/2022	Issued for Interdisciplinary Check
R01	22/07/2022	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

Table of Contents

Row	24" Line Pipe Material Data Sheet		
1		Project:	FEED for Gas Gathering Pipelines & Associated Infrastructure-Phase 1
2		Field name:	BBOU Manifold Stattion to ELEP Manifold Stattion .
3		Geographical zone:	South Nigeria
4		Line Identification:	24" Pipeline from BBOU to ELEP.
5	References: 1) DEP 31.20.40.37-Gen "Linepipe For Critical Service. 2) ISO 3183: 2007 Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems. 3) EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-001 Specification for Line Pipes		
6	Data Sheet Units		Include units with value (e.g. kg or lbs, m or ft etc)
7	Datasheet Key:	Purchaser to Provide	
8	Service application		
9		Location:	Onshore · Land and Swamp
10		Fluid type:	Gas and Condensate · If multiphase, specify liquid-to-gas ratic
11		Fluid service: Region 0	P_{H2S} (bar): · P_{CO2} (bar): · Ph:
12	EA-BF	Fluid chemical composition:	
13		Design temperature (°C)	Min.: · Max.: 60
14		Pressure (Bar)	External: · Internal: 110
15		Total length:	**8700 m**
16		Pipelay method:	Open Cut / HDD
17	API SPEC 5L - Confirmation of applicability of individual NORMATIVE Annexes		
18		Annex B (normative) Manufacturing procedure qualification for PSL 2 pipe	Applicable
19		Annex C (normative) Treatment of surface imperfections and defects	Applicable
20		Annex D (normative) Repair welding procedure	Applicable
21		Annex E (normative) Non-destructive Inspection for Other than Annex H, J, and/or	Applicable
22		Annex F (normative) Requirements for couplings (PSL 1 only)	Not Applicable
23		Annex G (normative) PSL 2 pipe with resistance to ductile fracture propagation	Applicable
24		Annex H (normative) PSL 2 pipe ordered for sour service	Not Applicable
25		Annex I (normative) Pipe ordered as "Through the Flowline" (TFL) pipe	Not Applicable
26		Annex J (normative) PSL 2 pipe ordered for offshore service	Not Applicable
27		Annex K (normative) Non-destructive Inspection for Pipe Ordered for Sour Service, Offshore Service, Fatigue Service and/or Service Requiring Longitudinal Plastic Strain Capacity	Applicable
28		Annex M (informative) Specification for Welded Jointers	Applicable
29		Annex N (Normative) PSL 2 Pipe Ordered for Applications Requiring Longitudinal Plastic Strain Capacity	Not Applicable
30	API Monogram		
31		API Monogram required: Yes	
32	Applicable Design Code(s)		
33		API RP 1111	Not applicable
34		ASME B31.4	Not applicable
35		ASME B31.8	Applicable
36		ISO 13623	Applicable
37		NEN 3650	Applicable
38		DNVGL-ST-F101: If applicable state supplementary requirements:	Not applicable
39		with Supplementary requirements D (enhanced dimensional)	Not applicable
40		with Supplementary requirements F (fracture arrest properties)	Not applicable
41		with Supplementary requirements P (plastic deformation)	Not applicable
42		with Supplementary requirements S (sour service)	Not applicable
43		with Supplementary requirements U (high utilisation)	Not applicable
44		Other:	
45	Surveillance by second party		
46		Surveillance by second or third party: Note: If Manufacturer is requested to provide a 3.2 Certificate , this can be achieved by either Manufacturer appointing an independent inspection authority, or the Company providing a third party inspector who acts as independant inspection authority to endorse all certificate for 3.2 certifcate.	Manufacturer uses in house inspection for 3.1 certificate, Purchaser provides third Party inspector to verify manufacturing.
47	Pipe Characteristics		
48		Pipe manufacture specification :	API 5L
49		Nominal Pipe diameter (inch)	24"
50		Outside diameter (mm)	610
51		Wall Thickness (mm)	15.88mm
52		Steel grade:	L483 / X70
53		Delivery condition:	R
54		Pipe manufacturing method :	SMLS
55		Pipe length (ft / meters):	40/12
56		Pipe length range : (ft /meters)	35-45/10.67-13.72
57		Type of length (ref API 9.11.3.3):	Double Random
58			% of the order is acceptable with the range of · meters
59		Mother Pipe For Bends required :	R
60		Note: 1. Assume pipe wt reduces by 8% on induction bending. 2. If mother pipe for bends has different specification/grade than main pipe then a new LPDMS is required for motherpipe for bends.	R
61		Motherpipe for bend Wall Thickness (inch /mm) before bending.	R
62		End preparation requirements:	Bevelled To API requirements
63		Bevel specific requirements	Bevelled To API requirements
64		Coating Required, system , thickness and specification.	Other 3LPE 3.0mm DEP 31.40.20.37 Gen
65		Miniumum Coating cut back requirement.	152mm
66	Starting Material		
67	8.3.3	Test on slab - Segregation / Inclusion content test required	Yes · Frequency: · first & last heat
68	8.3.5	Shearing for HFW & SAWH /COWH hot rolled coil allowed (instead of machining) :	No
69	Specific Pipe Testing requirement		

70	Chemistry	Chemistry - Additional requirements:	Not applicable
71	Table 18 item 4	Elevated temperature tensile test required:	Yes
72		Test frequency:	Production test (and MPQT)
73	9.3.2	Speciments type to be tested:	Longitudinal tensile testing
74			Longitudinal tensile testing
75			Longitudinal tensile testing
76			Longitudinal tensile testing
77		Acceptance criteria (Ys ; Mpa)	
78		Test temperature (°C):	
79	Table 18 item 3a	Longitudinal tensile test required:	Yes
80		If required, test Frequency:	MPQT only
81	9.3.2	Transverse & longitudinal testing restriction on Yield /Tensile Rt0,5/Rm ratio requirement, if different from S-616 clause 9.3.2.	≤ 0.93%
82	Table 20 note c	Use of transverse test pieces for tensile tests of SMLS pipe instead of Longitudinal as specified in API 5L table 20.	Yes
83	Table 20 (API 5L) , footnote c	Use of the ring expansion test for transverse yield strength determinations:	Acceptable
84	API 10.2.4.3	International standard applicable to Charpy testing: notch 2mm depth 0.25mm radius	ISO 148-1
85	9.8.2	Alternative Charpy test temperature scheme: standard is Table 8 of S-616.	
86	9.8.2.2	Pipe body shear fracture area required: (Shear area on Charpy specimen)	Yes
87	9.8.3	Pipe HAZ shear fracture area required: (Shear area on charpy in HAZ)	Yes
88	Table 18 item 10	DWT test required: (Only valid for 406 mm OD (16 inch) and up in gas service)	Yes
89	9.9.1	If yes, test temperature:	
90	Table 18 item 12	CTOD test required:	No
91		If yes, test temperature:	
92	9.16	Acceptance criteria:	Material Certificate
93	9.17	Spring back test to be performed:	No
94	Table 18 item 6 9.19	Macro & micro Base metal examination required:	Yes
95	Table 18 item 6c and 6d	Hardness testing required:	Yes
96	10.2.5.5 & H.7.3.3.1	Indentation scheme figures 14, 15 & 16 Annex J1 Figure 16 change C to 1mm.	Annex J
97	Table 18 item 23 9.13.3	Pipe ends etching required for misalignment of weld beads in SAWL & COWL	Yes
98	**Surface conditions, imperfections and defects**		
99	9.10.3.2	Arc burns repair non dressable not allowed :	Not Allowed
100	**Dimensional**		
101	9.11.3.1	Alternative dimensional tolerances: options J3 or S-616 section J.6.1 table J9 A or B	
102	9.12.5.6	Internal machining required: Refer DEP 31.40.20.37 Appendix 1.	No Length:
103	9.13.2.2	OD weld to be ground down:	No Length:
104	9.13.4	2.5mm peaking agreed for SAWH welds:	No
105	API Table 10 footnote b	Diameter andout-of-roundness tolerances tolerances for the ends of seamless pipe with t > 0.984 inches (25.0 mm).	Yes
106	API 10.2.8.1	Specific method to be used for determining pipe diameter (ID or OD measurement).	Yes
107	**Inspection**		
108	8.3.10	Certificate 3.2 for plate and coil is required. Either Manufacturer has independent inspection , or Company Third Party Inspection performs the independnant inspection .	
109	API 10.1.2.1	Certificate 3.2 for pipe is required: As stated above.	Yes
110	Table 18, footnote c	Alternative test unit definition required:	
111		Test unit is defined as a pipe lot coming from the same size, same heat number and consist of: Max	Pipes for diameter
112		Requirement to limit test unit to 50 pipes if design temperature is below -10°C applicable:	
113	**Hydrotest**		
114	10.2.6.5	Hydrotest as per API 5L 46th ed. Section 10.2.6.7 of API, 95% yield based on min wall thickness.	API 5L
115	API 10.2.6.7	Use of minimum permissible wall thickness to determine hydrostatic test pressure.	Yes
116	**Retesting**		
117	10.2.12	Increased retesting requirement.	Yes
118	**Marking, pipe protection**		
119	14	Bevel protectors required:	Yes
120		Type:	
121	11.2.5	Varnish type coating required:	Yes
122	11.2.5	Varnish type coating only over lettering	Yes
123	API 11.2.4	Round low stress die stamp (or vibro etching) required on plain end or weld bevel	Yes
124	Marking requirements		
125		*Purchaser should inform Manufacturer of any specific requirements for: colored painted band on pipe, additional marking information (ref API 11.2.1), second side marking (full or reduced) . For the marking: distance from pipe end (consider 200mm from pipe end), specified location (ID/OD), lettering size.*	

126	**Documentation**							
127	10.1.3.1	Final reports / certificate format: searchable PDF, data curves required in native forma						
128	S-616L	Production Histograms required for individual projects.	Yes					
129		Photos of macro and micro required.	Yes					
130	10.1.3.1	Number of originals and hard copies of Mill certificate and final production report:	2 hard copies and a PDF file of all records.					
131	13	Manufacturing data and inspection records to be kept for a period of: 10 years						
132	10.1.3.1	Native datas required for tests in which data curves are developed.	Yes					
133	**Annex B - Manufacturing Procedure Qualification for PSL 2 Pipe**							
134	B.5.2	Manufacture Procedure qualification: MPQT , production prior to all test results at mills	MPQT					
146	E.3.1.3	For seamless pipe, requirement for NDE to be performed after hydrotest:	Applicable					
147	Personnel Qualification							
148	E.1.1	Qualification of personnel performing NDT subject to Purchaser Approval:	Yes					
149		Personnel certification scheme to be as per: ISO or ASNT or alternative approved by						
150	EMI							
151	Table E.2	EMI testing required:						
152	Table E.1	Full-body inspection using the flux leakage method allowed:						
153		Full-body inspection using the eddy current method allowed:						
154	Pipe end / Bevel inspection							
155	E.3.2.3 & E.3.3.2 & API K.2.1.3	Pipe end UT lamination check - longer distance of examination required: min 150mm (6 inch).	Yes	150mm				
156	E.3.2.3 & E.3.3.2	Pipe end UT lamination check by MUT: AUT or Semi AUT for pipe end UT.	Yes					
157	E.3.2.5 & E.3.3.3 K.2.1.4	MPI of end face or pipe bevel required:	Yes					
158	Radiographic testing							
159	E.3.1.1	Digital Radiography is acceptable subject review of system as part of mill audit.	Yes					
160	API E.4.3.1	Alternate IQI type required:	No	If yes, type:				
161	Ultrasonic Testing							
162	E.3.1.1	Pipe weld seam inspection by RT and MUT allowed:	Yes					
163	E.5.2.2	Alternative length for reference standards used for dynamic demonstration. Full pipe lengths for calibration pipe required.	Yes	50mm				
164	E.5.1.1.1	Capability of oblique defects detection required: Yes but only for seamless pipe	Yes					
165	E.5.1.1.1	Systems that record defect positioning with inspection maps are acceptable: Only with prior review and approval of system.	Yes					
166	Table E.7.3 note h	Oblique angle to be determined using notched billet.	No					
167	E.5.1.1.2	Plate / Coil coverage requirements for lamination testing by UT. 100% coverage capable .	Applicable					
168	E.9 & Table E.7.1	Detection of laminar imperfections by UT to be performed on both plate and along weld seam:	Yes					
169	E.11 & K.4.3	UT to be used to verify that the pipe body is free (within the designated fault limitations) of laminar imperfections.	Yes					
170	Table E.7.1 & Table E.7.2 & Table E.7.3	Alternative N5 notch length (12,5 or 25 mm) : 50mm notch length is acceptable.	Yes	50mm				
171	Table E.7.1 note 9 & Table E.7.2 note 12	Lamination check to be performed using a 6mm (ISO size) diameter, standard is 0.25 inch or 6.4mm FBH:	Yes					
172	E.5.2.3	Alternative or additional UT reference reflectors required	No					
173	Table E.8.1 note	Acceptance criteria level as per Table K.1:	Level 1					
174	E.5.9	MUT inspection for Delayed Hydrogen Cracking (DHC) detection:	Not Applicable					
175	E.5.9.2	Alternative probe arrangement for DHC detection (if so state alternative):	Not Acceptable					
176	Residual magnetism							
177	E.7.5	Residual magnetism check required:	Yes					
178	Wall thickness inspection							
179	E.12.1	Seamless pipe wall thickness check 100% coverage.	Yes					
180	E.12.2.1	Plate wall thickness check required.	Yes					
181	E.12.2.2	Welded pipe wall thickness check for pipe , if plate or hot rolled coil thickness standard deviation in mill exceeds 0.25mm.	Yes					
182	Additional Requirements Specific to Annex K							
183	K.3.2.1 & K.3.2.2	Pipe Body or strip/plate body acceptance criteria: as per S-616 Table K1.	Level 1					
184	K.2.1.1	Laminar imperfection at pipe end limited to 3.2mm in any direction.	Applicable					
185	K.4.3	Alternative distance from the edge or the edge of the longitudinal weld seam for laminar imperfection testing.	No	if yes, length:				
186		Acceptance criteria for edge / weld seam edge laminar imperfection. Min level 2	Level 2					
187	API K.3.3	Increased coverage for ultrasonic thickness measurements for SMLS pipe:	Yes	to	≥		%	25
188	Table E.2 & K.3.4.4	Full-body magnetic particle inspection of Seamless pipe required: MPQT only:	Yes					
189	API K.5.4	Magnetic particle inspection of the weld seam at the pipe ends of SAW pipe required:	No					

190	**Annex G - PSL 2 Pipe with Resistance to Ductile Fracture Propagation**			
191	G.1	Additional pipe body charpy required: Refer pipeline design DEP 31.40.00.10-Gen.	Yes	
192	G.1	Approach selected by Purcahser:	Approach 1	
193	G.6	Acceptance criteria:	Material Certificate	
194	**Annex H - PSL 2 Pipe Ordered for Sour Service**			
195	Table H.3 item 8	Alternative test frequency:	No	If yes, test frequency:
196	H.7.2.2	Increased sampling: testing of three sets of three specimen instead of three specimen.	No	
197	H.4.3	CLR, CTR and CSR limit to be applied to each section if different from S-616	No	
239	N.4.2.2	Alternative maximum yield strength requirement	No	if yes, specify:
240	N.4.2.2	Alternative maximum Tensile strength requirement	No	if yes, specify:
241	N.4.2.2	Alternative maximum ratio of YS/TS in longitudinal direction requirement	No	if yes, specify:
242	N.8.3.1	Type of longitudinal tensile specimen:		
243	N.8.3.1	Stress-strain curve to be produced with an alternative frequency:	No	if yes, specify:
244	API N.4.2.4 & API N.4.2.5	Stress-strain curve shape requirement to be applied:		
245	N.4.5	Alternative CTOD acceptance criteria to be applied: SCR 0.5mm.	No	if yes, specify:
246	Table N.6	Increased testing frequency for HFW pipes:	No	if yes, specify:
247	**Appendix 1 - Welding Consumables**			
248	AP1.1	Weld deposit containing more than 1 % mass fraction nickel	Acceptable	
249	**Appendix 2 : Weldability Test**			
250	AP 4.3.2.1	Butt welds - Construction / fabrication code: WPS required for review and approval		
251	9.15	Weldability test required:	Yes	
252	9.15	If no, historical Weldability data required:	Yes	
253	9.15	Historical Datas: endorsment by third party required	Yes	
254	9.15	sPWHT testing required:	Yes	
255	9.15	PWHT temperature/time range recommendation required:	Yes	
262	**ISO 3183 - Annex A Applicability**			
263		Annex A PSL 2 pipe ordered for European onshore natural gas transmission pipelines	Not Applicable	
264	**Confirmation of applicability of individual Appendixes**			
265		Appendix 1 Welding Consumables	Applicable	
266		Appendix 2 Weldability Test	Applicable	
267		Appendix 3 Qualification of NDT at plate/coil and pipe mill	Applicable	
268		Appendix 4 NDT procedure minimum content	Applicable	

36" Line Pipe Material Data Sheet

Row							
1		Project:	FEED for Gas Gathering Pipelines & Associated Infrastructure-Phase 1				
2		Field name:	ELEPA Manifold Stattion to AKABEL Pigging Stattion .				
3		Geographical zone:	South Nigeria				
4		Line Identification:	36" Pipeline from BBOU to ELEP.				
5	References: 1) DEP 31.20.40.37-Gen "Linepipe For Critical Service. 2) ISO 3183: 2007 Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems. 3) EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-001 Specification for Line Pipes						
6	Data Sheet Units		Include units with value (e.g. kg or lbs, m or ft etc)				
7	Datasheet Key:	Purchaser to Provide					
8	**Service application**						
9		Location:	Onshore	Land and Swamp			
10		Fluid type:	Gas and Condensate	If multiphase, specify liquid-to-gas ratic			
11		Fluid service:	Region 0	P_{H2S} (bar):		P_{CO2} (bar):	Ph:
12	EA-BF	Fluid chemical composition:					
13		Design temperature (°C)	Min.:		Max.:	60	
14		Pressure (Bar)	External:		Internal:	110	
15		Total length:	32000 m				
16		Pipelay method:	Open Cut / HDD				
17	**API SPEC 5L - Confirmation of applicability of individual NORMATIVE Annexes**						
18		Annex B (normative) Manufacturing procedure qualification for PSL 2 pipe	Applicable				
19		Annex C (normative) Treatment of surface imperfections and defects	Applicable				
20		Annex D (normative) Repair welding procedure	Applicable				
21		Annex E (normative) Non-destructive Inspection for Other than Annex H, J, and/or	Applicable				
22		Annex F (normative) Requirements for couplings (PSL 1 only)	Not Applicable				
23		Annex G (normative) PSL 2 pipe with resistance to ductile fracture propagation	Applicable				
24		Annex H (normative) PSL 2 pipe ordered for sour service	Not Applicable				
25		Annex I (normative) Pipe ordered as "Through the Flowline" (TFL) pipe	Not Applicable				
26		Annex J (normative) PSL 2 pipe ordered for offshore service	Not Applicable				
27		Annex K (normative) Non-destructive Inspection for Pipe Ordered for Sour Service, Offshore Service, Fatigue Service and/or Service Requiring Longitudinal Plastic Strain Capacity	Applicable				
28		Annex M (informative) Specification for Welded Jointers	Applicable				
29		Annex N (Normative) PSL 2 Pipe Ordered for Applications Requiring Longitudinal Plastic Strain Capacity	Not Applicable				
30	**API Monogram**						
31		API Monogram required:	Yes				
32	**Applicable Design Code(s)**						
33		API RP 1111	Not applicable				
34		ASME B31.4	Not applicable				
35		ASME B31.8	Applicable				
36		ISO 13623	Applicable				
37		NEN 3650	Applicable				
38		DNVGL-ST-F101: If applicable state supplementary requirements:	Not applicable				
39		with Supplementary requirements D (enhanced dimensional)	Not applicable				
40		with Supplementary requirements F (fracture arrest properties)	Not applicable				
41		with Supplementary requirements P (plastic deformation)	Not applicable				
42		with Supplementary requirements S (sour service)	Not applicable				
43		with Supplementary requirements U (high utilisation)	Not applicable				
44		Other:					
45	**Surveillance by second party**						
46		Surveillance by second or third party: Note: If Manufacturer is requested to provide a 3.2 Certificate , this can be achieved by either Manufacturer appointing an independent inspection authority, or the Company providing a third party inspector who acts as independant inspection authority to endorse all certificate for 3.2 certifcate.	Manufacturer uses in house inspection for 3.1 certificate, Purchaser provides third Party inspector to verify manufacturing.				
47	**Pipe Characteristics**						
48		Pipe manufacture specification :	API 5L				
49		Nominal Pipe diameter (inch)	36"				
50		Outside diameter (mm)	914				
51		Wall Thickness (mm)	23.83mm				
52		Steel grade:	L483 / X70				
53		Delivery condition:	R				
54		Pipe manufacturing method :	SAWL				
55		Pipe length (ft / meters):	40/12				
56		Pipe length range : (ft /meters)	35-45/10.67-13.72				
57		Type of length (ref API 9.11.3.3):	Double Random				
58				% of the order is acceptable with the range of		meters	
59		Mother Pipe For Bends required :	R				
60		Note: 1. Assume pipe wt reduces by 8% on induction bending. 2. If mother pipe for bends has different specification/grade than main pipe then a new LPDMS is required for motherpipe for bends.	R				
61		Motherpipe for bend Wall Thickness (inch /mm) before bending.	R				
62		End preparation requirements:	Bevelled To API requirements				
63		Bevel specific requirements	Bevelled To API requirements				
64		Coating Required, system , thickness and specification.	Other 3LPE	3.0mm	DEP 31.40.20.37 Gen		
65		Miniumum Coating cut back requirement.	152mm				
66	**Starting Material**						
67	8.3.3	Test on slab - Segregation / Inclusion content test required	Yes	Frequency:	first & last heat		
68	8.3.5	Shearing for HFW & SAWH /COWH hot rolled coil allowed (instead of machining) :	No				
69	**Specific Pipe Testing requirement**						

70	Chemistry	Chemistry - Additional requirements:	Not applicable
71	Table 18 item 4	Elevated temperature tensile test required:	Yes
72		Test frequency:	Production test (and MPQT)
73	9.3.2	Speciments type to be tested:	Longitudinal tensile testing
74			Longitudinal tensile testing
75			Longitudinal tensile testing
76			Longitudinal tensile testing
77		Acceptance criteria (Ys ; Mpa)	
78		Test temperature (°C):	
79	Table 18 item 3a	Longitudinal tensile test required:	Yes
80		If required, test Frequency:	MPQT only
81	9.3.2	Transverse & longitudinal testing restriction on Yield /Tensile Rt0,5/Rm ratio requirement, if different from S-616 clause 9.3.2.	≤ 0.93%
82	Table 20 note c	Use of transverse test pieces for tensile tests of SMLS pipe instead of Longitudinal as specified in API 5L table 20.	Yes
83	Table 20 (API 5L) , footnote c	Use of the ring expansion test for transverse yield strength determinations:	Acceptable
84	API 10.2.4.3	International standard applicable to Charpy testing: notch 2mm depth 0.25mm radius	ISO 148-1
85	9.8.2	Alternative Charpy test temperature scheme: standard is Table 8 of S-616.	
86	9.8.2.2	Pipe body shear fracture area required: (Shear area on Charpy specimen)	Yes
87	9.8.3	Pipe HAZ shear fracture area required: (Shear area on charpy in HAZ)	Yes
88	Table 18 item 10	DWT test required: (Only valid for 406 mm OD (16 inch) and up in gas service)	Yes
89	9.9.1	If yes, test temperature:	
90	Table 18 item 12	CTOD test required:	No
91		If yes, test temperature:	
92	9.16	Acceptance criteria:	Material Certificate
93	9.17	Spring back test to be performed:	No
94	Table 18 item 6 9.19	Macro & micro Base metal examination required:	Yes
95	Table 18 item 6c and 6d	Hardness testing required:	Yes
96	10.2.5.5 & H.7.3.3.1	Indentation scheme figures 14, 15 & 16 Annex J1 Figure 16 change C to 1mm.	Annex J
97	Table 18 item 23 9.13.3	Pipe ends etching required for misalignment of weld beads in SAWL & COWL	Yes
98	**Surface conditions, imperfections and defects**		
99	9.10.3.2	Arc burns repair non dressable not allowed :	Not Allowed
100	**Dimensional**		
101	9.11.3.1	Alternative dimensional tolerances: options J3 or S-616 section J.6.1 table J9 A or B	
102	9.12.5.6	Internal machining required: Refer DEP 31.40.20.37 Appendix 1.	No Length:
103	9.13.2.2	OD weld to be ground down:	No Length:
104	9.13.4	2.5mm peaking agreed for SAWH welds:	No
105	API Table 10 footnote b	Diameter andout-of-roundness tolerances tolerances for the ends of seamless pipe with t > 0.984 inches (25.0 mm).	Yes
106	API 10.2.8.1	Specific method to be used for determining pipe diameter (ID or OD measurement).	Yes
107	**Inspection**		
108	8.3.10	Certificate 3.2 for plate and coil is required. Either Manufacturer has independent inspection , or Company Third Party Inspection performs the independnant inspection .	
109	API 10.1.2.1	Certificate 3.2 for pipe is required: As stated above.	Yes
110	Table 18, footnote c	Alternative test unit definition required:	
111		Test unit is defined as a pipe lot coming from the same size, same heat number and consist of:	Max Pipes for diameter
112		Requirement to limit test unit to 50 pipes if design temperature is below -10°C applicable:	
113	**Hydrotest**		
114	10.2.6.5	Hydrotest as per API 5L 46th ed. Section 10.2.6.7 of API, 95% yield based on min wall thickness.	API 5L
115	API 10.2.6.7	Use of minimum permissible wall thickness to determine hydrostatic test pressure.	Yes
116	**Retesting**		
117	10.2.12	Increased retesting requirement.	Yes
118	**Marking, pipe protection**		
119	14	Bevel protectors required:	Yes
120		Type:	
121	11.2.5	Varnish type coating required:	Yes
122	11.2.5	Varnish type coating only over lettering	Yes
123	API 11.2.4	Round low stress die stamp (or vibro etching) required on plain end or weld bevel	Yes
124	Marking requirements		
125		*Purchaser should inform Manufacturer of any specific requirements for: colored painted band on pipe, additional marking information (ref API 11.2.1), second side marking (full or reduced) . For the marking: distance from pipe end (consider 200mm from pipe end), specified location (ID/OD), lettering size.*	

126		**Documentation**						
127	10.1.3.1	Final reports / certificate format: searchable PDF, data curves required in native forma						
128	S-616L	Production Histograms required for individual projects.	Yes					
129		Photos of macro and micro required.	Yes					
130	10.1.3.1	Number of originals and hard copies of Mill certificate and final production report:	2 hard copies and a PDF file of all records.					
131	13	Manufacturing data and inspection records to be kept for a period of: 10 years						
132	10.1.3.1	Native datas required for tests in which data curves are developed.	Yes					
133		**Annex B - Manufacturing Procedure Qualification for PSL 2 Pipe**						
134	B.5.2	Manufacture Procedure qualification: MPQT , production prior to all test results at mills risk.	MPQT					
146	E.3.1.3	For seamless pipe, requirement for NDE to be performed after hydrotest:	Applicable					
147		Personnel Qualification						
148	E.1.1	Qualification of personnel performing NDT subject to Purchaser Approval:	Yes					
149		Personnel certification scheme to be as per: ISO or ASNT or alternative approved by Company.						
150		EMI						
151	Table E.2	EMI testing required:						
152	Table E.1	Full-body inspection using the flux leakage method allowed:						
153		Full-body inspection using the eddy current method allowed:						
154		Pipe end / Bevel inspection						
155	E.3.2.3 & E.3.3.2 & API K.2.1.3	Pipe end UT lamination check - longer distance of examination required: min 150mm (6 inch).	Yes	150mm				
156	E.3.2.3 & E.3.3.2	Pipe end UT lamination check by MUT: AUT or Semi AUT for pipe end UT.	Yes					
157	E.3.2.5 & E.3.3.3 K.2.1.4	MPI of end face or pipe bevel required:	Yes					
158		Radiographic testing						
159	E.3.1.1	Digital Radiography is acceptable subject review of system as part of mill audit.	Yes					
160	API E.4.3.1	Alternate IQI type required:	No	If yes, type:				
161		Ultrasonic Testing						
162	E.3.1.1	Pipe weld seam inspection by RT and MUT allowed:	Yes					
163	E.5.2.2	Alternative length for reference standards used for dynamic demonstration. Full pipe lengths for calibration pipe required.	Yes	50mm				
164	E.5.1.1.1	Capability of oblique defects detection required: Yes but only for seamless pipe	Yes					
165	E.5.1.1.1	Systems that record defect positioning with inspection maps are acceptable: Only with prior review and approval of system.	Yes					
166	Table E.7.3 note h	Oblique angle to be determined using notched billet.	No					
167	E.5.1.1.2	Plate / Coil coverage requirements for lamination testing by UT. 100% coverage capable .	Applicable					
168	E.9 & Table E.7.1	Detection of laminar imperfections by UT to be performed on both plate and along weld seam:	Yes					
169	E.11 & K.4.3	UT to be used to verify that the pipe body is free (within the designated fault limitations) of laminar imperfections.	Yes					
170	Table E.7.1 & Table E.7.2 & Table E.7.3	Alternative N5 notch length (12,5 or 25 mm) : 50mm notch length is acceptable.	Yes	50mm				
171	Table E.7.1 note 9 & Table E.7.2 note 12	Lamination check to be performed using a 6mm (ISO size) diameter, standard is 0.25 inch or 6.4mm FBH:	Yes					
172	E.5.2.3	Alternative or additional UT reference reflectors required	No					
173	Table E.8.1 note	Acceptance criteria level as per Table K.1:	Level 1					
174	E.5.9	MUT inspection for Delayed Hydrogen Cracking (DHC) detection:	Not Applicable					
175	E.5.9.2	Alternative probe arrangement for DHC detection (if so state alternative):	Not Acceptable					
176		Residual magnetism						
177	E.7.5	Residual magnetism check required:	Yes					
178		Wall thickness inspection						
179	E.12.1	Seamless pipe wall thickness check 100% coverage.	Yes					
180	E.12.2.1	Plate wall thickness check required.	Yes					
181	E.12.2.2	Welded pipe wall thickness check for pipe , if plate or hot rolled coil thickness standard deviation in mill exceeds 0.25mm.	Yes					
182		Additional Requirements Specific to Annex K						
183	K.3.2.1 & K.3.2.2	Pipe Body or strip/plate body acceptance criteria: as per S-616 Table K1.	Level 1					
184	K.2.1.1	Laminar imperfection at pipe end limited to 3.2mm in any direction.	Applicable					
185	K.4.3	Alternative distance from the edge or the edge of the longitudinal weld seam for laminar imperfection testing.	No	if yes, length:				
186		Acceptance criteria for edge / weld seam edge laminar imperfection. Min level 2	Level 2					
187	API K.3.3	Increased coverage for ultrasonic thickness measurements for SMLS pipe:	Yes	to	≥		%	25
188	Table E.2 & K.3.4.4	Full-body magnetic particle inspection of Seamless pipe required: MPQT only:	Yes					
189	API K.5.4	Magnetic particle inspection of the weld seam at the pipe ends of SAW pipe required:	No					

190	**Annex G - PSL 2 Pipe with Resistance to Ductile Fracture Propagation**			
191	G.1	Additional pipe body charpy required: Refer pipeline design DEP 31.40.00.10-Gen.	Yes	
192	G.1	Approach selected by Purcahser:	Approach 1	
193	G.6	Acceptance criteria:	Material Certificate	
194	**Annex H - PSL 2 Pipe Ordered for Sour Service**			
195	Table H.3 item 8	Alternative test frequency:	No	If yes, test frequency:
196	H.7.2.2	Increased sampling: testing of three sets of three specimen instead of three specimen.	No	
197	H.4.3	CLR, CTR and CSR limit to be applied to each section if different from S-616	No	
239	N.4.2.2	Alternative maximum yield strength requirement	No	if yes, specify:
240		Alternative maximum Tensile strength requirement	No	if yes, specify:
241		Alternative maximum ratio of YS/TS in longitudinal direction requirement	No	if yes, specify:
242	N.8.3.1	Type of longitudinal tensile specimen:		
243	N.8.3.1	Stress-strain curve to be produced with an alternative frequency:	No	if yes, specify:
244	API N.4.2.4 & API N.4.2.5	Stress-strain curve shape requirement to be applied:		
245	N.4.5	Alternative CTOD acceptance criteria to be applied: SCR 0.5mm.	No	if yes, specify:
246	Table N.6	Increased testing frequency for HFW pipes:	No	if yes, specify:
247	**Appendix 1 - Welding Consumables**			
248	AP1.1	Weld deposit containing more than 1 % mass fraction nickel	Acceptable	
249	**Appendix 2 : Weldability Test**			
250	AP 4.3.2.1	Butt welds - Construction / fabrication code: WPS required for review and approval		
251	9.15	Weldability test required:	Yes	
252		If no, historical Weldability data required:	Yes	
253		Historical Datas: endorsment by third party required	Yes	
254		sPWHT testing required:	Yes	
255		PWHT temperature/time range recommendation required:	Yes	
262	**ISO 3183 - Annex A Applicability**			
263		Annex A PSL 2 pipe ordered for European onshore natural gas transmission pipelines	Not Applicable	
264	**Confirmation of applicability of individual Appendixes**			
265		Appendix 1 Welding Consumables	Applicable	
266		Appendix 2 Weldability Test	Applicable	
267		Appendix 3 Qualification of NDT at plate/coil and pipe mill	Applicable	
268		Appendix 4 NDT procedure minimum content	Applicable	